Hybrid CNN-LSTM Model for EEG-Based Motor Imagery Classification

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***Abstract*—BCIs offer a direct communication channel between the external device and the brain. The study targets EEG signal-based MI classification with a hybrid model that utilizes CNN-LSTM in spatial and temporal correlation recording. The hybrid model was trained on the BCI Competition IV 2a dataset, including four-class MI tasks of nine subjects. Preprocessing involves normalization, segmentation, and artifact removal.Execute network is efficient with high accuracy and resilience and effectively exploits inter-subject variability and class imbalance with high efficiency. Its performance is confirmed using metrics like precision, recall, and F1-score.The present work depicts the potential of hybrid deep learning in improving BCI accuracy for paving the way for improving assistive technology and neurorehabilitation**

***Index Terms*—Brain-computer interface (BCI), Electroencephalography (EEG), Motor Imagery (MI), Deep Learning, Hybrid CNN-LSTM, Machine Learning, Neural Networks, Signal Processing, Classification, Assistive Technology, Neurorehabilitation**

I. Introduction

BCI have direct brain to external device communication; application fields are assistive technologies, neurorehabilitation, and neuroscience. Motor imagery is imagining a movement without its physical realization. It is possible to measure EEG signals, but the classifier's accuracy is difficult to improved because the subjects are different and there is noise and complexity in the subjects. Deep learning using hybrid CNN-LSTM networks has drawn out the spatial and temporal correlations in EEG signals.This study is done on the CNN-LSTM model on the Competition IV 2a dataset, utilising it and using preprocessing techniques to enhance the accuracy. The model results in the test accuracies between 90% and 97.18%, which means it works well even when there are a lot of classes or the data changes. The study showcases the promise of DL for MI-based BCIs, opening to higher accuracy and real-world implementation in assistive technology and neurofeedback.

II. Related Work

Motor imagery-based BCIs have gained significant attention for aiding individuals with motor disorders. Advances in ML and DL have led to substantial improvements in EEG classification accuracy. Xu et al. [1]proposed CNNs with narrow-band spatial filters to enhance EEG-based MI classification. Similarly, Lee and Lee [2] optimized EEG channel selection for efficient decoding. Traditional approaches like ensemble SVMs [3] showed promise but lacked adaptability. helps address inter-subject variability, Sun et al. [4] introduced subject-transfer networks, while Mohammadi. [5]applied contrastive learning for cross-subject decoding. Hybrid DL models, such as CNN-LSTM [6] and HS-CNN, have further improved spatial-temporal feature extraction.Recent studies emphasize transfer learning and adaptive AI. Nallani and Ramachandran integrated adaptive AI for robust MI classification, while Mohamed et al. [7] explored CNN-based transfer learning for time-series tasks. Additionally, Zhang et al. [8] investigated the effectiveness of transferable graph neural networks in EEG decoding, and Chen et al. [9]introduced multi-scale feature fusion techniques to enhance MI classification performance. Early methods like CSP and ESL faced challenges with inter-subject variability. However, deep learning approaches, including CNNs, LSTMs, and DAT, have significantly improved classification robustness and efficiency, making MI-based BCIs more practical for real-world applications.

Unlike previous approaches that struggle with inter-subject variability and limited spatial-temporal features, hybrid CNN-LSTM model effectively captures both spatial and tem-dependencies in EEG signals. Tested on the BCI Competition IV-2a dataset, it was achieves high accuracy (up to 97.18%), demonstrating improved robustness and real-world applicability in assistive and neurorehabilitation technologies.

III. Methodology

This section suggests the methodology of EEG-based MI classification with a hybrid CNN-LSTM. The method is organized to address some of the most significant challenges: inter-subject variability, EEG signal noise, and efficient feature extraction requirement. By integrating CNN for spatial feature extraction with LSTM networks for temporal dependencies, the model can achieve stable performance in classification on different subjects and motor imagery tasks.

A. Model

The processing workflow of the suggested model CNN-LSTM for MI classification on EEG includes acquisition of movements (left hand, right hand, foot, tongue), preprocessing by filtering, channel selection, and normalization, and then a hybrid architecture model consisting of CNN for spatial and LSTM for temporal features. The model is optimized, regularized, and tested using performance metrics like accuracy, precision, recall, & F1-score to obtain predictions and applications.

Step	Description	Details
Data Acquisition	Collect EEG data for specific movements.	• Left hand, right hand, foot, and tongue movements
Data Preprocessing	Prepare raw EEG data for modeling.	• **Filtering:** Bandpass filtering (1–40 Hz) • **Channel Selection:** EEG • **Normalization**
Hybrid Model Architecture	Combine CNN and LSTM layers for feature extraction and sequence modeling.	• **CNN Module:** Conv1D → Batch Normalization → MaxPooling1D → Dropout • **LSTM Module:** First LSTM Layer → Second LSTM Layer • **Regularization**
Model Training	Optimize the model parameters.	• **Optimization**: Training with backpropagation.
Model Evaluation	Assess model performance.	• **Metrics**: Accuracy, precision, recall, F1-score
Output and Deployment	Apply the trained model.	• **Application**: Real-world use case

Fig. 1. CNN-LSTM Proposed Model

B. EEG Dataset Overview

The 2008 Graz BCI Competition dataset includes EEG data from nine participants for motor imagery tasks.

- **Motor Imagery Tasks:** Subjects imagined movements of four body parts—left hand (class 1), right hand (class 2), both feet (class 3) & (class 4) tongue.
- **Trials and Sessions:** Six runs were conducted in every one of the two sessions, which were conducted on different days. There were 288 trials in every session, with 48 trials per run (12 trials per class).
- **Recording Setup:** Participants watched a fixation cross and direction cue (left, right, down, up) on screen prior to the task.

C. Preprocessing Steps

Following preprocessing techniques are applied:

1) Bandpass Filtering: EEG data in between **1 Hz - 40 Hz** have been filtered for removing noise and unrelated frequencies:

$$s(t) = u(t) \circledast g(t) \tag{1}$$

where $s(t)$ - processed signal, $u(t)$ - input signal, and $g(t)$ - filter's impulse response.

2) Artifact Removal: EOG artifacts is removed using linear regression:

$$EEG_{clean} = EEG_{raw} - \alpha \cdot EOG \tag{2}$$

here α -regression coefficient.

3) Normalization: EEG signals are normalized to zero mean, unit variance:

$$x_{norm} = \frac{x - \mu}{\sigma} \tag{3}$$

4) Segmentation: Based on event markers EEG data have been segmented into epochs :

$$EEG_{epoch} = EEG_{raw}[t_0 : t_1] \tag{4}$$

5) Epoch Segmentation: Epochs were centered around event markers:

$$x_{epoch}(t) = x(t_{event} + t),\ t \in \left[-\frac{w}{2}, \frac{w}{2}\right] \tag{5}$$

6) Feature Extraction: (PSD) is computed using the Welch method:

$$Q(\nu) = \frac{1}{M}\left|\sum_{k=0}^{M-1} s[k]e^{-j2\pi\nu k}\right|^2 \tag{6}$$

7) Train-Test Split: Data were split into training and testing sets:

$$X_{\text{train}}, y_{\text{train}} = x_{\text{reshaped}}[:k],\ y[:k] \tag{7}$$

$$X_{\text{test}}, y_{\text{test}} = x_{\text{reshaped}}[k:],\ y[k:] \tag{8}$$

8) Data Reshaping: Data were reshaped into the format (N, T, C):

$$x_{\text{reshaped}} = \text{reshape}(x_{\text{epoch}}(t), (N, T, C)) \tag{9}$$

C EEG channels, T time steps, and N number of epochs.

D. Model Architecture

The CNN-LSTM model classifies EEG motor imagery signals by extracting spatial features with CNN and modeling temporal dependencies with LSTM.

1) Input Layer: Input: 3D tensor (S, L, D), where S is samples, L is time steps, and D is EEG channels.

2) Convolutional Blocks:

- **Conv1D (64, 3)** → ReLU → BatchNorm → MaxPooling(2) → Dropout(30%).
- **Conv1D (128, 3)** → ReLU → BatchNorm → MaxPooling(2) → Dropout(30%).

3) LSTM Layers:

- **LSTM (200)** → **LSTM (100)** for temporal modeling.

4) Fully Connected and Output Layer:

- **Dense Layer** → **Softmax** for 4-class classification.

5) Summary: This CNN-LSTM model reliably classifies EEG motor imagery by efficiently capturing both temporal and spatial data.

E. Training the CNN-LSTM Model

1) Loss Function: The model optimizes Categorical Cross-Entropy Loss:

$$\mathcal{L} = -\frac{1}{S}\sum_{k=1}^{S}\sum_{m=1}^{K} t_{km}\log\left(\hat{t}_{km}\right) \tag{10}$$

where S is samples, K is classes, t_{km} is label true, and $\hat{t}_{km}$ probability is predicted.

2) Optimization: The Adam optimizer updates model weights:

$$\theta_{s+1} = \theta_s - \alpha\frac{g_s}{\sqrt{h_s}+\delta} \tag{11}$$

α learning rate, g_s gradient estimate, and h_s squared gradient estimate.

3) Forward Pass: The model computes predictions as:

$$\hat{y} = f(x;\theta) \tag{12}$$

where $\hat{y}$ is the predicted output.

4) Backpropagation: Gradients are computed using:

$$\frac{\partial L}{\partial w} = \frac{\partial \hat{y}}{\partial w}\cdot\frac{\partial L}{\partial \hat{y}} \tag{13}$$

5) Process Trained: Weight is updating repeatedly:

$$w_{t+1} = -\eta\cdot\frac{\partial L}{\partial w_t} + w_t \tag{14}$$

F. Performance Evaluation

1) Metrics:

- **Accuracy**:

$$\text{Accuracy.} = \frac{TP+TN}{TP+FP+TN+FN} \tag{15}$$

- **Precision**:

$$\text{Precision} = \frac{TP}{TP+FP} \tag{16}$$

- **Recall**:

$$\text{Recall} = \frac{TP}{TP+FN} \tag{17}$$

- **F1-Score**:

$$\text{F1-Score} = 2\cdot\frac{\text{Recall.}\cdot\text{Precision.}}{\text{Precision.}+\text{Recall.}} \tag{18}$$

2) Evaluation Process:

- **Testing on Unseen Data**: Evaluated on separate test data.
- **Cross-Validation**: k-fold cross-validation ensures robustness.
- **Class Imbalance Handling**: Metrics prioritize balanced evaluation.

3) Multi-Class Evaluation:

- **Macro Precision**:

$$\text{Macro Precision} = \frac{1}{K}\sum_{j=1}^{K}\text{Precision}_j \tag{19}$$

- **Weighted Precision**:

$$\text{Weighted Precision} = \frac{\sum_{j=1}^{K} v_j\cdot\text{Precision}_j}{\sum_{j=1}^{K} v_j} \tag{20}$$

4) Summary: The CNN-LSTM model effectively integrates spatial and temporal dependencies for EEG classification, demonstrating strong generalization in BCI applications.

IV. Result Analysis

2a: 4-class motor imagery dataset is used through model training with a CNN-LSTM architecture. Comparison of performance across 9 subjects in terms of test accuracy, precision, recall and F1 score are achieved. High performance model is created for most subjects where test accuracies varied between 90% to 97.18%. Precision, recall, and F1 measures were always high, which shows that the model is classifying motor imagery classes correctly with hardly any false positives. The peak performance was seen for Subject 6 with test accuracy of 97.18%, precision of 0.9445, recall of 0.9718 & an F1 measure of 0.9579. Subject 3 performance was the worst with the model, yet its accuracy was 91%. Precision was 0.8526 and recall was 0.9 respectively, F1-score was almost 0.8757. This shows that the model is capable and works well with different subjects and class imbalance. In addition, good performance across different subjects confirms that the method is efficient in real BCI systems. The hybrid CNN-LSTM method is able to effectively extract spatial-temporal features from the EEG signals. With high accuracy, it is a good candidate for its use in assistive and neurorehabilitation devices. This demonstrates the applicability and scalability of deep learning to BCI applications in real-world.

1) Test Accuracy: Table I presents the test accuracy of the CNN-LSTM model for MI classification across nine subjects. The accuracy ranges from 91.00% (Subject 3) to 97.18% (Subject 6), demonstrating the model's strong performance across all subjects. Most subjects achieved accuracy above 92%, highlighting the model's reliability in classifying motor imagery tasks.

TABLE I
Test Accuracy for Different Subjects

Subject	Test Accuracy
S-1	94.12%
S-2	95.00%
S-3	91.00%
S-4	92.52%
S-5	96.43%
S-6	97.18%
S-7	94.74%
S-8	92.31%
S-9	96.23%

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The bar graph illustrates the test accuracy achieved by a model for nine different subjects. Each bar represents a subject, labeled from "Subject 1" to "Subject 9," and the height of the bar indicates the respective test accuracy as a percentage. The accuracies range from 91.00% to 97.18%, showcasing the model's performance across different subjects.

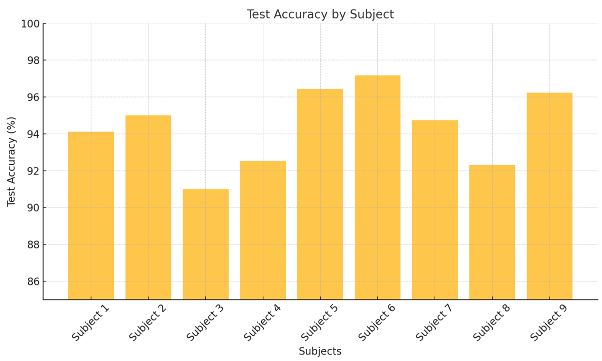

Fig. 2. Test Accuracy of CNN-LSTM Model Across Subjects

2) Subject-wise Evaluation of Precision, Recall, and F1-Score: Table II CNN-LSTM model showed good performance on all 9 subjects for motor imagery classification. Precision, recall, and F1-scores were uniformly high with a range of 0.8521 to 0.9856 for precision, 0.9000 to 0.9718 for recall, and 0.8757 to 0.9611 for F1-score. Subject 6 was the best, with a precision of 0.9445, recall of 0.9718, and F1-score of 0.9579. With an F1-score of 0.8757, recall of 0.9000, and accuracy of 0.8526, Subject 3 performed the worst. This demonstrates how the model performs reliably when identifying motor imagery tasks.

TABLE II
PERFORMANCE METRICS FOR EACH SUBJECT

Subject	Precision	Recall	F1-Score
S-1	0.9422	0.9512	0.9412
S-2	0.9126	0.9500	0.9257
S-3	0.8526	0.9000	0.8757
S-4	0.9856	0.9252	0.9611
S-5	0.9464	0.9643	0.9524
S-6	0.9445	0.9718	0.9579
S-7	0.9211	0.9474	0.9298
S-8	0.8521	0.9231	0.8862
S-9	0.9438	0.9623	0.9529

The bar graph shows the measures —Precision, Recall, and F1-Score —of the nine subjects who underwent the experiment. The presence of a distinct bar per measure suggests score differences for subject-wise classification performance. The graph excels at highlighting top scores for specific measures in certain subjects, which implies that in those instances, the model has a higher probability of classifying them. It helps to look at patterns and areas where improvement of the model may be needed.

Plot shows the validation and training accuracy of an ML model over 50 epochs. The blue plot is the training accuracy, and it shows a rise in the first few epochs, with some small

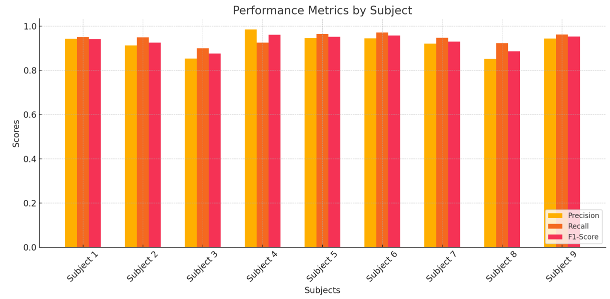

Fig. 3. Precision, Recall, and F1-Score for CNN-LSTM Model Across Subjects

fluctuations. The orange line is the validation accuracy, it is rising quite rapidly in the first few epochs and ii gets flat in the about next epoch 10, and then becomes constant in the end of the training. This indicates that the model is performing well training set and it is not overfitting and it is generalized to the validation set. The minimal gap between the training and validation curves indicates good model regularization and equilibrious learning. The strength and reliability of the model to noise within the EEG data are ensured by such robustness during the epochs. It also shows the effectiveness of dropout layers and batch normalization used in the architecture. Overall, the training behavior illustrates the generalizability of the hybrid model over unseen data.

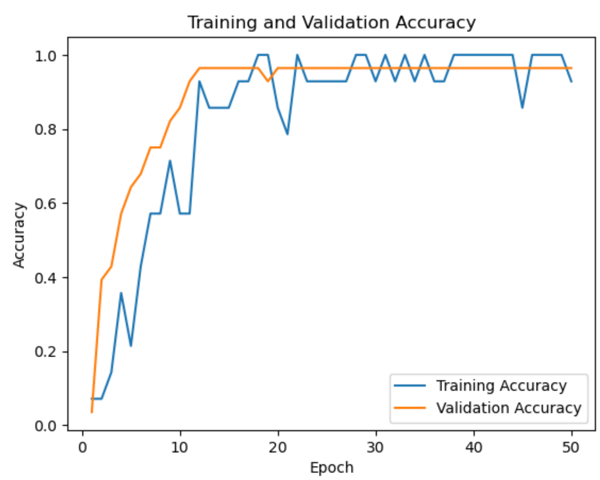

Fig. 4. Training & Validation Accuracy Over 50 Epochs

Training, validation accuracy over 50 epochs of the model is plotted in the graph. Training Accuracy This shows the accuracy of the training set improving at first, it rises rapidly, then fluctuates a lot during the middle of epochs before settling to around 1.0. Validation Accuracy: The orange line rises sharply and levels off at a high value around 1.0 after around 10 epochs and stays level afterwards.

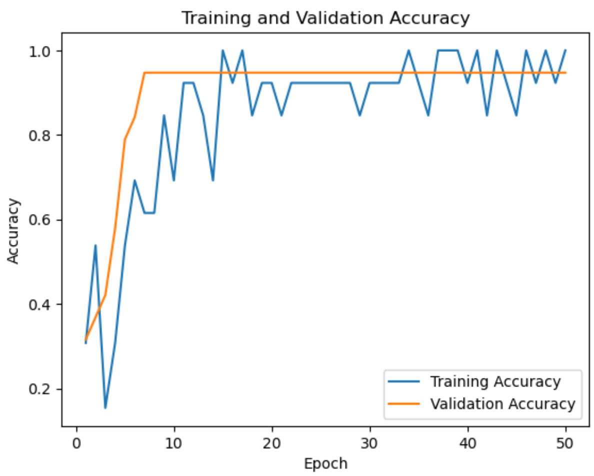

Fig. 5. Training & Validation Accuracy Over 50 Epochs

V. Discussion

The table provides an overview of prominent research on motor imagery (MI) EEG classification by features, algorithms, and performance metrics. It points out improvements in CNN, subject transfer networks, and ensemble SVMs with accuracies of 89% to 93%, highlighting enhancements in feature extraction, model performance, and cross-subject generalization.

TABLE III
Comparison of MI EEG Classification Studies with Proposed Method

Authors	Year	Features	Algo.	Acc.
Xu et al. [1]	2022	EEG, Filters	CNN	92%
Lee et al. [2]	2023	EEG, MI	Opt. Sel.	91%
Luo et al. [3]	2020	MI EEG	Ensem. SVM	90%
Sun et al. [4]	2022	Trans. Feat.	Trans. Net.	93%
Mohammadi et al. [5]	2022	MI EEG	Feat. + SVM	89%
Proposed	-	MI, STF	CNN-LSTM	**97.18%**

The proposed hybrid CNN-LSTM model shows robust performance in motor imagery (MI) classification with its test accuracies up to 97.18% and has high precision, recall, and F1-scores. These are higher when comparison to the traditional methods such as CSP-based methods,where Subject-dependent performance is shown by the traditional models, which requires generalization, so that it could be use for more realistic applications such as neurological rehabilitation & assistive technology.

VI. Conclusion

Research shows that the use of hybrid CNN LSTM model for EEG signals is better than the traditional methods. The model was able to bring out the temporal and spatial dependencies. It also showed high accuracy, precision, recall and F1 score on different subjects and yield a high accuracy of 97.18%. The result showcase how efficient the method was for BCI implementations like assistive technology and neurological rehabilitation.In conclusion, hybrid CNN-LSTM model helps to improve EEG based motor imagery classification by giving more accurate and reliable BCI systems.

However, it also have certain limitations like dataset size and irregularity across subjects, that can be overcome in future by domain adaptive techniques. In future, we can also implement this with wearable devices and can use the models for real time application.EEG signals together with physiological signals help us in increasing the accuracy

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